



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 01

- 1. Title:** Demonstrate inline, internal, and external JavaScript, use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), Web Browser (Google Chrome / Microsoft Edge), HTML5, JavaScript, Windows Operating System.
- 3. Theory:** JavaScript is a client-side scripting language used to make web pages interactive and dynamic. It can be added to a webpage in three ways:
 - **Inline JavaScript:** Code written directly inside HTML elements using event attributes.
 - **Internal JavaScript:** Code written inside the `<script>` tag within the HTML file.
 - **External JavaScript:** Code written in a separate .js file and linked to the HTML document.
 - In this experiment, JavaScript is used to display a welcome message, show user information, and print output using console methods such as `console.log()`, `console.warn()`, and `console.error()`.



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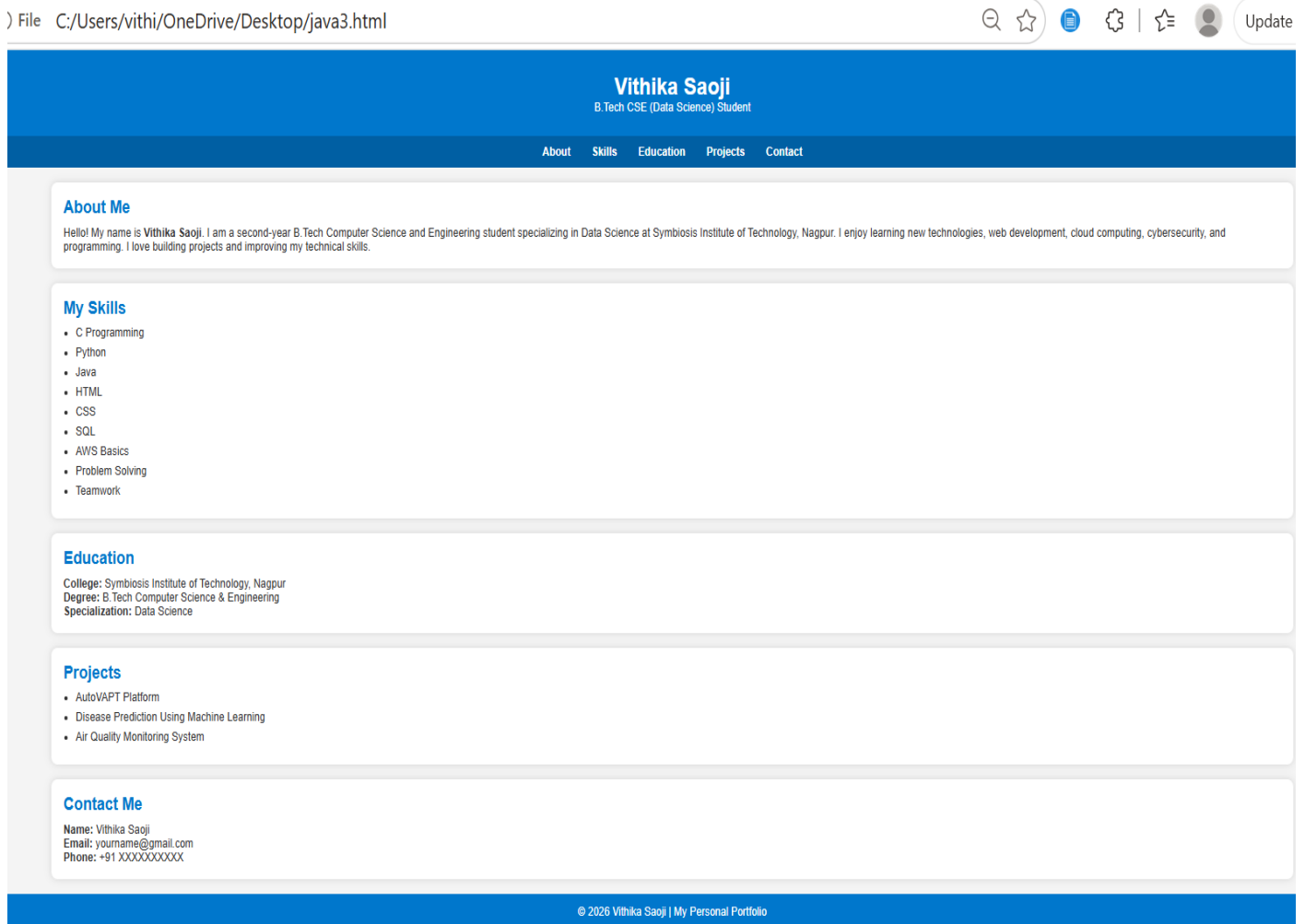
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4. Output:



5. Result/Conclusion: The webpage was successfully created using HTML and JavaScript. It demonstrated the use of inline, internal, and external JavaScript. User information and a welcome message were displayed correctly, and console methods produced the expected output in the browser's developer console. This experiment helped understand different ways of implementing JavaScript and using console methods for debugging.



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Case Study No.:

1. Case Study Title: Create a webpage for college department and link the student information page with it and write a java script to display “welcome to sit nagpur” in an alert box.

2. Case Study Program Code:

```
File Edit Selection View Go Run Terminal Help
index.html JS script.js JAVA.html student_info.html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>SIT Nagpur - Department of Computer Science</title>
7   <style>
8     /* Internal CSS */
9     body { font-family: Arial, sans-serif; margin: 0; padding: 0; background-color: #f4f7f6; }
10    header { background-color: #004085; color: white; padding: 20px; text-align: center; }
11    nav { background-color: #005cbf; padding: 10px; text-align: center; }
12    nav a { color: white; margin: 0 15px; text-decoration: none; font-weight: bold; }
13    nav a:hover { text-decoration: underline; }
14    .container { padding: 20px; text-align: center; }
15    button { padding: 10px 20px; font-size: 16px; margin: 10px; cursor: pointer; background-color: #28a745; color: white; border: none; border-radius: 5px; }
16    button:hover { background-color: #218838; }
17  </style>
18 </head>
19 <body>
20
21   <header>
22     <h1>Welcome to Department of IT</h1>
23     <h2>SIT NAGPUR</h2>
24   </header>
25
26   <nav>
27     <a href="index.html">Home</a>
28     <a href="student_info.html">Student Information</a>
29   </nav>
30
31   <div class="container">
32     <h3>About Our Department</h3>
33     <p>Providing cutting-edge technology education and empowering future innovators at SIT Nagpur.</p>
34   </div>
35 </body>
36 </html>
```



```
File Edit Selection View Go Run Terminal Help

JS script.js C:\...\Desktop index1.html launch.json main.html index.html

JAVA.html > JS script.js > ...
1 // This function will be triggered by the inline button
2 function displayInlineAlert() {
3     alert("Welcome to SIT NAGPUR");
4 }
5
6 // Console logging for debugging using Browser Developer Tools
7 console.log("External JavaScript loaded successfully.");
8 console.log("Welcome to SIT NAGPUR - logged from the console!");
9
```

```
File Edit Selection View Go Run Terminal Help qwertyuiop

JS script.js C:\...\Desktop index1.html launch.json main.html index.html JS script.js JAVA.html student_info.html

JAVA.html > student_info.html > ...
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <title>Student Information - SIT NAGPUR</title>
6     <style>
7         body { font-family: Arial, sans-serif; padding: 20px; background-color: #f9f9f9; }
8         .container { max-width: 600px; margin: 0 auto; background: white; padding: 20px; box-shadow: 0 0 10px rgba(0,0,0,0.1); }
9         a { display: block; margin-top: 20px; text-decoration: none; color: #005cbf; }
10    </style>
11 </head>
12 <body>
13
14     <div class="container">
15         <h2>Student Information Portal</h2>
16         <p>Access student records, timetables, and academic performance data here.</p>
17
18         <form>
19             <label for="roll">Enter Student Roll No:</label><br><br>
20             <input type="text" id="roll" name="roll" required>
21             <button type="button" onclick="searchStudent()">Search Record</button>
22         </form>
23
24         <a href="index.html">Back to Home</a>
25     </div>
26
27     <script>
28         // Internal JavaScript specifically for this page
29         function searchStudent() {
30             let rollNo = document.getElementById('roll').value;
31             console.log("Searching database for Roll No: " + rollNo);
32             alert("Record search initiated for Roll No: " + rollNo);
33         }
34     </script>
35
```



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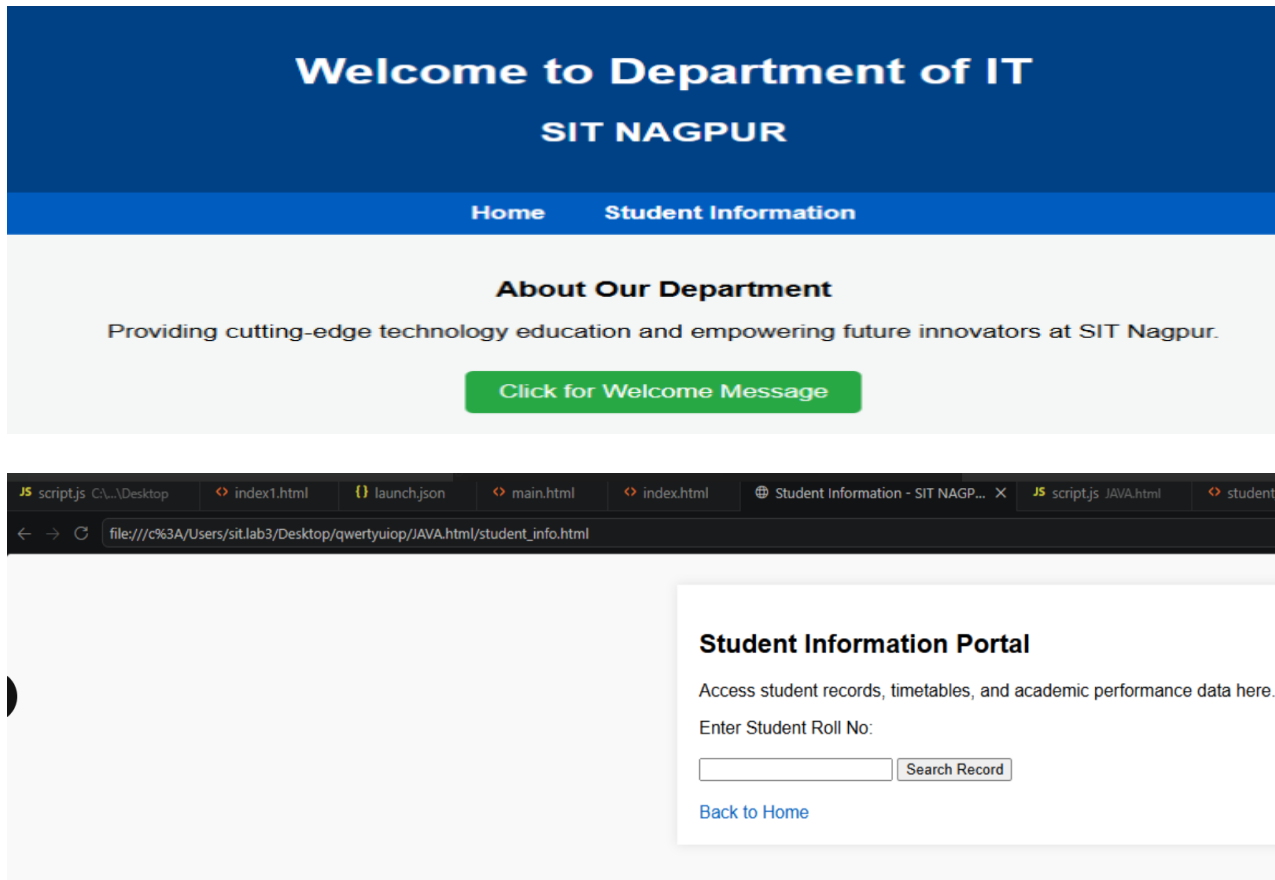
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3. Output:



4. Result/Conclusion: The college department webpage was created successfully. The Student Information page was linked to it, and JavaScript was implemented using inline, internal, and external methods. A welcome message was displayed in an alert box, and `console.log()` output was verified in the browser console.



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Experiment No.: 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `${}.`

The `document.write()` method displays the billing receipt on the webpage.

4. Output :

← × ⓘ File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

This page says

Enter Customer Name:

OK

Cancel

← × ⓘ File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

This page says

Enter Product Name:

OK

Cancel



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File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

This page says

Enter Quantity:

OK

Cancel

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

// Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Billing Calculator

Billing Receipt

Customer: Vithika

product: Laptop

Quantity: 3

Price per Item: ₹70000

GST (18%): ₹37800.00

Final Bill: ₹247800.00

5. Result/Conclusion: A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.:

1. Case Study Title: Develop the webpage for an online shopping website using HTML CSS and HTML5 tags.

2. Case Study Program Code:

```
Welcome JS Practical2.js index.html  
<> Case study 2 java.html > html > head > style > .container  
1 <!DOCTYPE html>  
2 <html lang="en">  
3 <head>  
4 <meta charset="UTF-8">  
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">  
6 <title>Online Shopping Bill</title>  
7  
8 <style>  
9 body{  
10     font-family:Arial,sans-serif;  
11     background: #f2f2f2;  
12 }  
13 .container{  
14     width:500px;  
15     margin:30px auto;  
16     background: white;  
17     padding:20px;  
18     border-radius:10px;  
19     box-shadow:0 0 10px gray;  
20 }
```

```
20 }  
21 h2{  
22     text-align:center;  
23     color: darkblue;  
24 }  
25 label{  
26     display:block;  
27     margin-top:10px;  
28     font-weight:bold;  
29 }  
30 input,select{  
31     width:100%;  
32     padding:8px;  
33     margin-top:5px;  
34 }  
35 button{  
36     width:100%;  
37     margin-top:20px;  
38     padding:10px;  
39     background: green;  
40     color: white;  
41     border:none;  
42     font-size:18px;  
43     cursor:pointer;  
44 }  
45 button:hover{  
46     background: darkgreen;  
47 }  
48 #result{
```

```

Case study 2.java.html > html > head > style
2  <html lang="en">
3  <head>
8  <style>
48  #result{
49      margin-top:20px;
50      padding:15px;
51      background: #eef;
52  }
53  </style>
54  </head>
55  <body>
56
57  <div class="container">
58
59  <h2>Online Shopping Bill</h2>
60
61  <label>Customer Name</label>
62  <input type="text" id="name">
63
64  <label>Mobile Number</label>
65  <input type="text" id="mobile">
66
67  <label>Invoice Number</label>
68  <input type="text" id="invoice">
69
70  <label>Product Name</label>
71  <input type="text" id="product">
72
73  <label>Quantity (Kg)</label>
74  <input type="number" id="qty">
75
76  <label>Rate per Kg (₹)</label>
77  <input type="number" id="rate">
78

```

```

Case study 2.java.html > html > body > div.container > button
2  <html lang="en">
55  <body>
57  <div class="container">
79  <label>Discount (%)</label>
80  <input type="number" id="discount">
81
82  <label>GST (%)</label>
83  <input type="number" id="gst">
84
85  <label>Packing Charges (₹)</label>
86  <input type="number" id="packing">
87
88  <label>Payment Mode</label>
89  <select id="payment">
90  <option>Cash</option>
91  <option>Card</option>
92  <option>UPI</option>
93  </select>
94
95  <label>Membership</label>
96  <select id="membership">
97  <option>Yes</option>
98  <option>No</option>
99  </select>
100
101  <button onclick="calculateBill()">Calculate Bill</button>
102  <div id="result"></div>
103  </div>
104  <script>
105
106  var shopName = "Fresh Mart";
107  let gstApplicable = true;
108  const country = "India";

```

```

Case study 2 java.html > html > body > script > calculateBill
2   <html lang="en">
55  <body>
104 <script>

109 let demoUndefined;
110 let demoNull = null;
111
112 function calculateBill(){
113
114     let name=document.getElementById("name").value;
115     let mobile=document.getElementById("mobile").value;
116     let invoice=document.getElementById("invoice").value;
117     let product=document.getElementById("product").value;
118     let qty=Number(document.getElementById("qty").value);
119     let rate=Number(document.getElementById("rate").value);
120     let discount=Number(document.getElementById("discount").value);
121     let gst=Number(document.getElementById("gst").value);
122     let packing=Number(document.getElementById("packing").value);
123     let payment=document.getElementById("payment").value;
124     let membership=document.getElementById("membership").value;
125     if(membership=="Yes"){
126         discount+=5;
127     }
128     let amount=qty*rate;
129     let discountAmount=(amount*discount)/100;
130     let afterDiscount=amount-discountAmount;
131     let gstAmount=(afterDiscount*gst)/100;
132     let finalBill=afterDiscount+gstAmount+packing;
133
134     document.getElementById("result").innerHTML=`
135
136     <h3>Invoice</h3>
137     <p><b>Shop Name:</b> ${shopName}</p>
138     <p><b>Customer Name:</b> ${name}</p>
139     <p><b>Mobile Number:</b> ${mobile}</p>
140     <p><b>Invoice Number:</b> ${invoice}</p>
141     <p><b>Product Name:</b> ${product}</p>
142     <p><b>Quantity:</b> ${qty} Kg</p>
143     <p><b>Rate:</b> ₹${rate}</p>
144     <p><b>Amount:</b> ₹${amount.toFixed(2)}</p>
145     <p><b>Discount:</b> ₹${discountAmount.toFixed(2)}</p>
146     <p><b>GST:</b> ₹${gstAmount.toFixed(2)}</p>

```

```

147     <p><b>Packing Charges:</b> ₹${packing.toFixed(2)}</p>
148     <p><b>Payment Mode:</b> ${payment}</p>
149     <p><b>Membership:</b> ${membership}</p>
150     <hr>
151
152     <h2>Total Bill : ₹${finalBill.toFixed(2)}</h2>
153     `;
154
155     console.log("Bill Generated");
156     console.info("Customer : "+name);
157     console.warn("GST Applied");
158     console.error("Demo Error Message");
159
160     console.log("Type Conversion:", Number("100"));
161     console.log("Type Coercion:", "100"+20);

```



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3. Output:

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Case%20study%202%20ja

Online Shopping Bill	
Customer Name	Vithika Saoji
Mobile Number	9756463728
Invoice Number	7657483928
Product Name	headphone
Quantity (Kg)	7
Rate per Kg (₹)	3500
Discount (%)	150
GST (%)	2
Packing Charges (₹)	50
Payment Mode	Cash
Membership	Yes
Calculate Bill	

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Case%20study%202%20java.html

Invoice
Shop Name: Fresh Mart
Customer Name: Vithika Saoji
Mobile Number: 9756463728
Invoice Number: 7657483928
Product Name: headphone
Quantity: 7 Kg
Rate: ₹3500
Amount: ₹24500.00
Discount: ₹37975.00
GST: ₹-269.50
Packing Charges: ₹50.00
Payment Mode: Cash
Membership: Yes
Total Bill : ₹-13694.50



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4. Result/Conclusion: The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. It accurately accepts customer and product details, calculates the final bill by applying discount, GST, and packing charges, and demonstrates the use of JavaScript data types, variable declarations, type conversion, and type coercion.



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Experiment No.: 03

1. Title: Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code), Web Browser (Google Chrome / Microsoft Edge), HTML, JavaScript, Windows Operating System.

3. Theory:

This program demonstrates the use of control structures and form validation in JavaScript. An HTML form is used to accept marks from the user. Validation checks whether the input is empty and ensures the entered marks are between 0 and 100. If the input is invalid, an alert message is displayed. If the input is valid, if-else control statements are used to determine the grade (A, B, C, D, or F) based on the entered marks, and the result is displayed on the webpage.

4. OUTPUT:



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File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Practical%203%20java%20

5. Result/Conclusion: The grading system was successfully implemented using HTML, CSS, and JavaScript. The program validates user input, prevents invalid entries, and accurately assigns grades based on the entered marks, demonstrating the effective use of form validation and control structures.



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Case Study No.: 03

1. Case Study Title: Write a JavaScript program that accepts student marks and displays the scholarship category.

2. Case Study Program Code:

```
JS Practical2.js  index.html  Case study 2 java.html  JS  II  ?  ↓  ↑  ↺  □

P3 case study.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Scholarship Eligibility</title>
7  <style>
8      body{
9          font-family: Arial, sans-serif;
10         background-color: #064c96;
11         display: flex;
12         justify-content: center;
13         align-items: center;
14         height: 100vh;
15     }
16     .container{
17         background: rgb(217, 4, 4);
18         padding: 30px;
19         width: 350px;
20         text-align: center;
21         border-radius: 10px;
22         box-shadow: 0 0 10px gray;
23     }
24     h1{
25         color: navy;
26     }
27     input{
28         width: 90%;
29         padding: 10px;
30         margin: 15px 0;
31         font-size: 16px;
32     }
```

```
33     button{
34         padding: 10px 20px;
35         background: navy;
36         color: white;
37         border: none;
38         border-radius: 5px;
39         cursor: pointer;
40         font-size: 16px;
41     }
42     button:hover{
43         background: blue;
44     }
45     #result{
46         margin-top: 20px;
47         font-size: 20px;
48         font-weight: bold;
49         color: green;
```



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```
51 </style>
52 </head>
53 <body>
54 <div class="container">
55 <h1>Vithika Saoji</h1>
56 <h2>Scholarship Eligibility</h2>
57 <input type="number" id="marks" placeholder="Enter Marks">
58 <br>
59 <button onclick="checkScholarship()">Submit</button>
60 <div id="result"></div>
61 </div>
62 <script>
63 function checkScholarship()
64 {
65     let marks = document.getElementById("marks").value;
66     if(marks=="")
67     {
68         alert("Enter Marks");
69         return;
70     }
71     marks = Number(marks);
72     if(marks<0 || marks>100)
73     {
74         alert("Enter Valid Marks");
75         return;
76     }
77     let scholarship;
78     if(marks<50)
79     {
80         scholarship = "Not Eligible";
81     }
82     else if(marks>=50 && marks<=69)
83     {
84         scholarship = "Bronze Scholarship";
85     }
86     else if(marks>=70 && marks<=84)
87     {
88         scholarship = "Silver Scholarship";
```

```
89     }
90     else if(marks>=85 && marks<=94)
91     {
92         scholarship = "Gold Scholarship";
93     }
94     else
95     {
96         scholarship = "Platinum Scholarship";
97     }
98     document.getElementById("result").innerHTML =
99     "Marks: " + marks + "<br>Scholarship: " + scholarship;
100 }
101 </script>
102 </body>
103 </html>
```

3. Output:

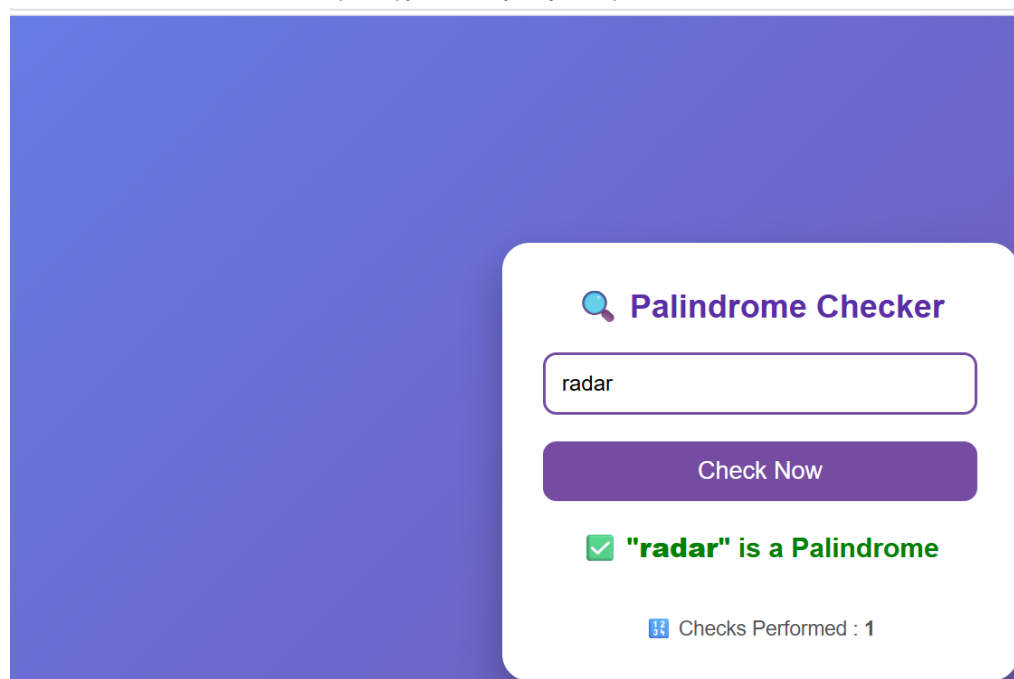
File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/P3%20case%20study.html

Experiment No.: 4

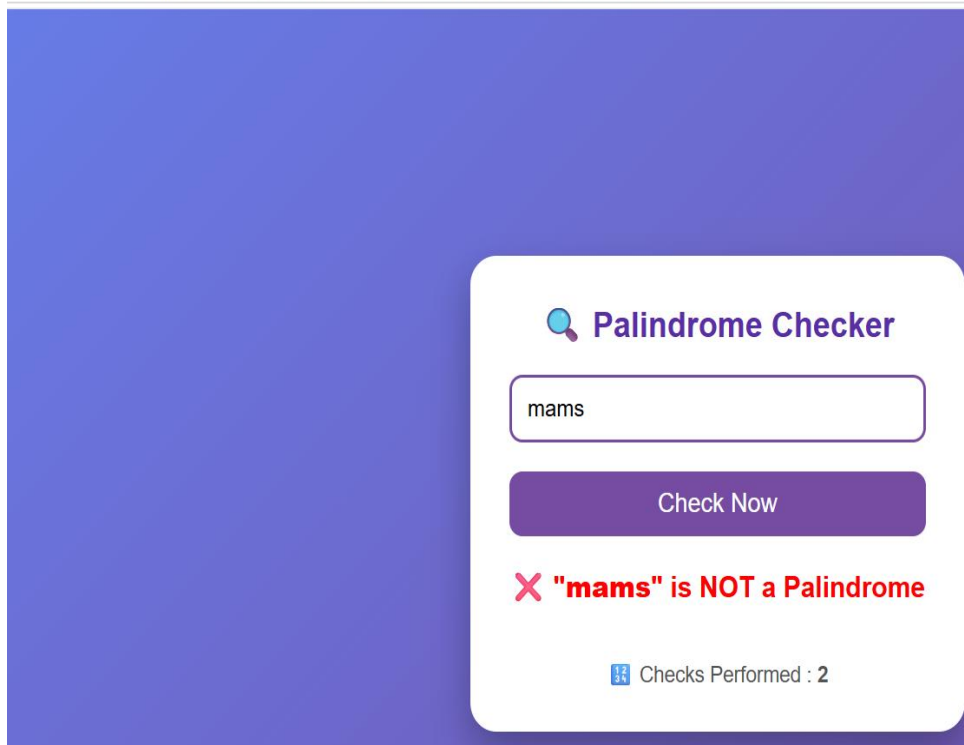
- 1. Title:** Use function types, scope, and closures; apply try catch; build a palindrome checker.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** This project is developed using HTML and JavaScript to create a Palindrome Checker. It demonstrates different JavaScript function types, including function declaration, function expression, and arrow functions. The program uses scope by defining global and local variables appropriately, while a closure is implemented to maintain and update the number of palindrome checks performed. Exception handling is achieved using the try...catch block, which prevents the program from crashing and displays meaningful error messages when the user enters invalid or empty input. The application accepts a word or phrase, removes unnecessary characters, checks whether it reads the same forwards and backwards, and displays the result through an interactive user interface.

4. Output :

① File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20Palindrome.html



) File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20Palindrome.html



 **Palindrome Checker**

mams

Check Now

✗ "mams" is NOT a Palindrome

 Checks Performed : 2

5. Result/Conclusion: The palindrome checker was successfully developed using HTML and JavaScript. It correctly checks whether the entered text is a palindrome, handles invalid input using try-catch, demonstrates function types, scope, and closures, and displays accurate results through a simple and interactive interface. This project enhanced the understanding of core JavaScript concepts while creating a reliable, user-friendly, and error-free web application.



Case Study No.: 04

1. Case Study Title: ATM Card PIN Verification Using JavaScript Functions.

2. Case Study Program Code:

```
Release Notes: 1.131.0 P4 Palindrome.html P4 casestudy.html X
P4 casestudy.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>ATM Card PIN Verification</title>
7
8 <style>
9 *{
10     margin:0;
11     padding:0;
12     box-sizing:border-box;
13     font-family:Arial, sans-serif;
14 }
15
16 body{
17     height:100vh;
18     display:flex;
19     justify-content:center;
20     align-items:center;
21     background:linear-gradient(135deg, #0f2027, #203a43, #2c5364);
22 }
23
24 .container{
25     width:360px;
26     background:#fff;
27     padding:30px;
28     border-radius:15px;
29     text-align:center;
30     box-shadow:0 0 20px rgba(0,0,0,0.3);
31 }
32
33 h2{
34     color:#2c5364;
35     margin-bottom:20px;
36 }
```

```
38 input{
39     width:100%;
40     padding:12px;
41     margin:15px 0;
42     border:2px solid #2c5364;
43     border-radius:8px;
44     font-size:18px;
45     text-align:center;
46 }
47
48 button{
49     width:100%;
50     padding:12px;
51     background:#2c5364;
52     color:white;
53     border:none;
54     border-radius:8px;
55     font-size:18px;
56     cursor:pointer;
57 }
58
59 button:hover{
60     background:#1b3d4f;
61 }
62
63 #result{
64     margin-top:20px;
65     font-size:18px;
66     font-weight:bold;
67 }
68
69 #attempts{
70     margin-top:10px;
71     color:#555;
72 }
73 </style>
74
```

```

74
75 </head>
76 <body>
77
78 <div class="container">
79
80 <h2>🏧 ATM PIN Verification</h2>
81
82 <input type="password" id="pin" maxlength="4" placeholder="Enter 4-Digit PIN">
83
84 <button onclick="verifyPIN()">Verify PIN</button>
85
86 <div id="result"></div>
87 <div id="attempts"></div>
88
89 </div>
90
91 <script>
92
93 // Global Scope
94 const correctPIN = "1234";
95
96 // Function Declaration
97 function getPIN(){
98     return document.getElementById("pin").value;
99 }
100
101 // Function Expression
102 const checkPIN = function(pin){
103     return pin === correctPIN;
104 };
105
106 // Closure
107 function loginCounter(){
108     let attempts = 0;
109

```

```

110         return function(){
111             attempts++;
112             return attempts;
113         }
114     }
115
116     const countAttempts = loginCounter();
117
118     // Arrow Function
119     const displayMessage = (message,color)=>{
120         document.getElementById("result").innerHTML = message;
121         document.getElementById("result").style.color = color;
122     };
123
124     // Main Function
125     function verifyPIN(){
126
127         try{
128
129             let pin = getPIN();    // Local Scope
130
131             if(pin===""){
132                 throw "Please enter your PIN.";
133             }
134
135             if(pin.length!=4 || isNaN(pin)){
136                 throw "PIN must contain exactly 4 digits.";
137             }
138
139             if(checkPIN(pin)){
140                 displayMessage("✅ PIN Verified Successfully!","green");
141             }
142             else{
143                 displayMessage("❌ Incorrect PIN. ","red");
144             }
145

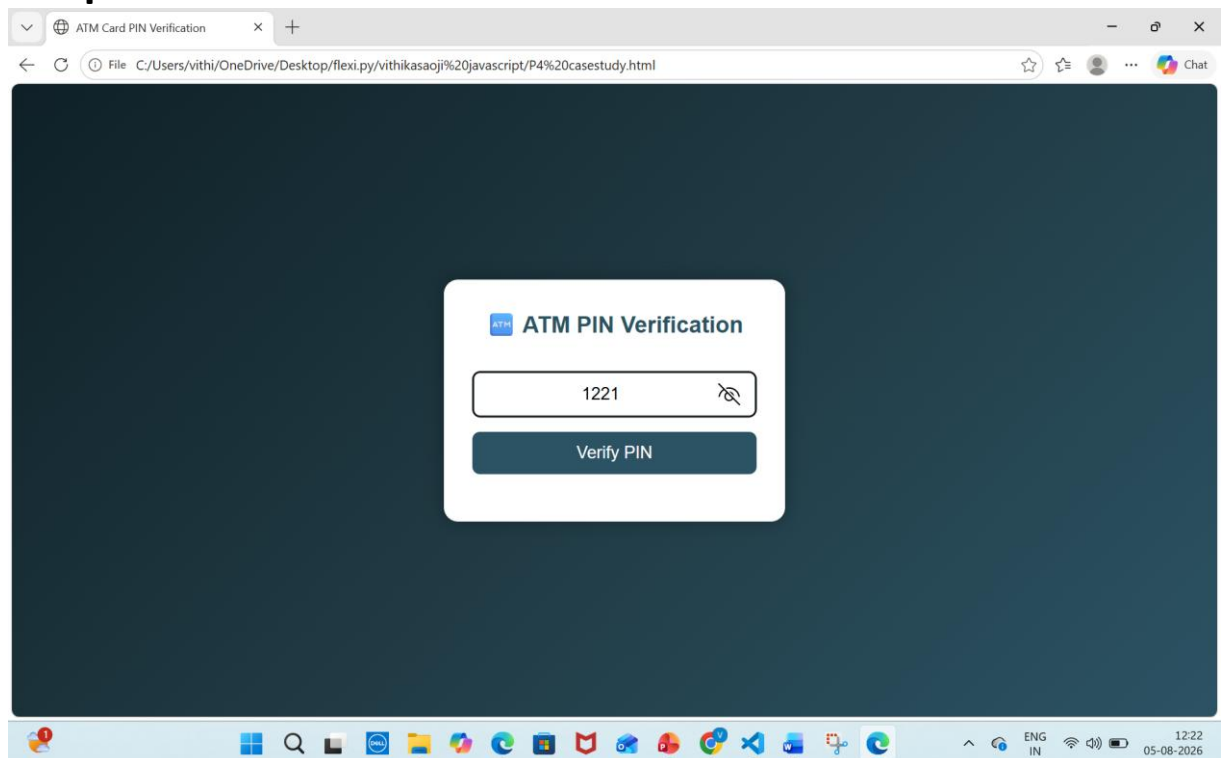
```

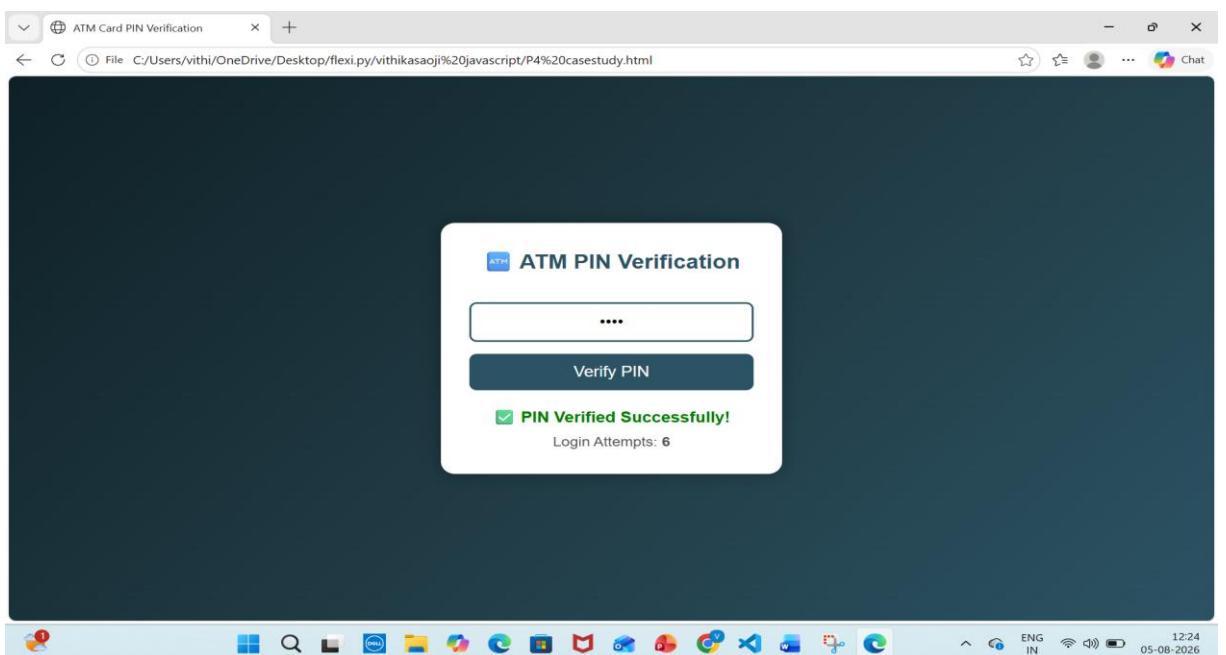
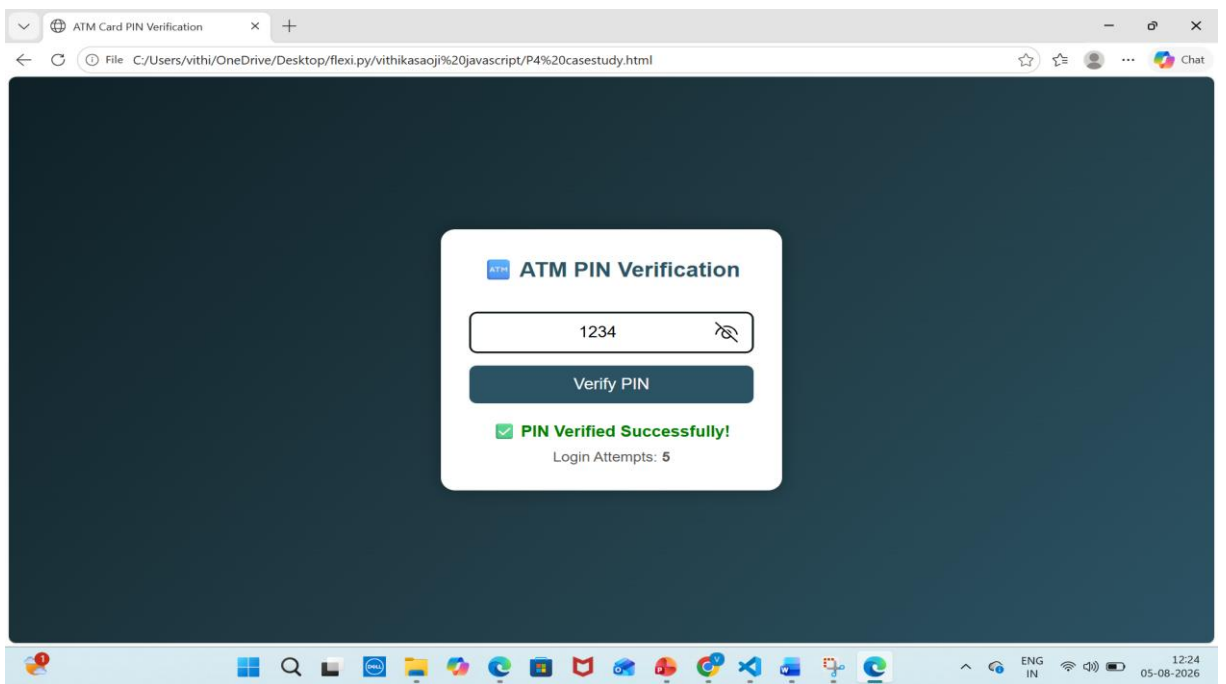
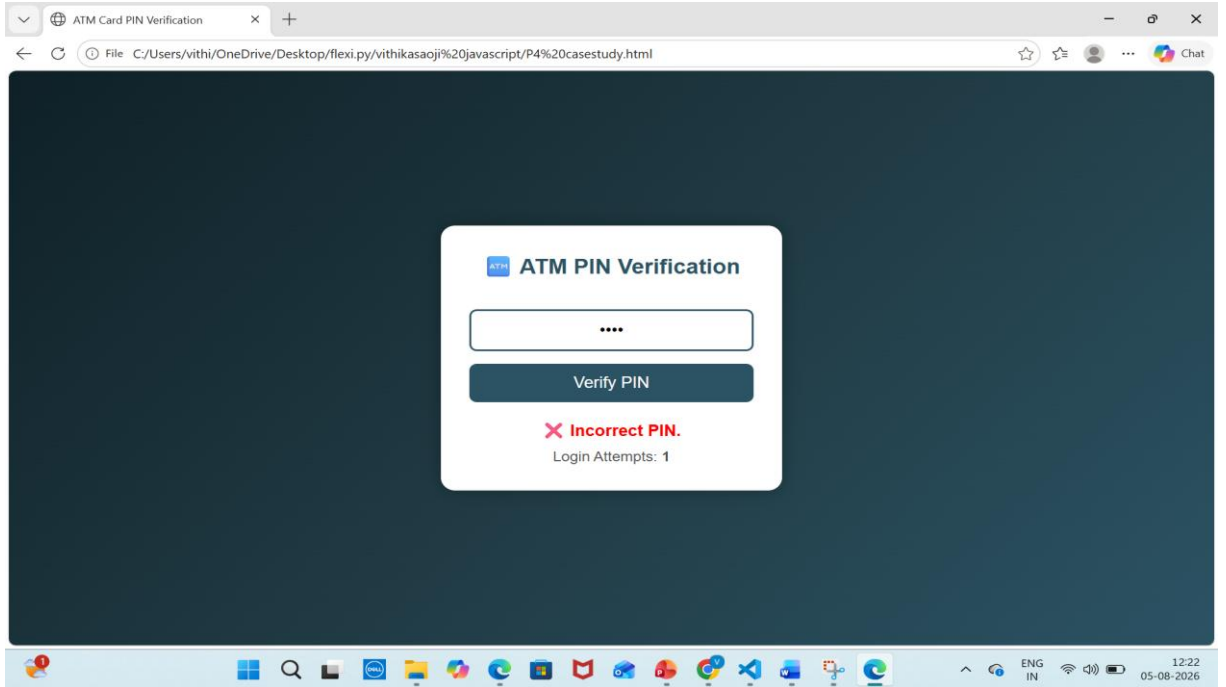
```

140     displayMessage("✅ PIN Verified Successfully.", "green");
141 }
142 else{
143     displayMessage("❌ Incorrect PIN.", "red");
144 }
145
146 document.getElementById("attempts").innerHTML =
147 "Login Attempts: <b>" + countAttempts() + "</b>";
148
149 }
150
151 catch(error){
152
153     displayMessage("⚠️ " + error, "orange");
154
155 }
156
157 finally{
158
159     console.log("PIN Verification Completed.");
160
161 }
162
163 }
164
165 </script>
166
167 </body>
168 </html>

```

3. Output:





4. Result & Conclusion: The ATM Card PIN Verification system was successfully implemented using HTML and JavaScript. It accurately verifies the entered PIN, handles invalid input using try-catch, demonstrates the use of function types, scope, and closures, and provides appropriate user feedback. This project helps in understanding core JavaScript concepts while creating a secure, reliable, and interactive PIN verification application.